

L11: Coupled Phase Oscillators – Kuramoto Model

Collective synchronization phenomena can be complex to model mathematically. The class of models that have been studied more extensively focuses on *coupled phase oscillators*.

Suppose that we have a system of N oscillators. The j 'th oscillator is a sinusoid function with angular frequency ω_j . To simplify suppose that the amplitude of all oscillators is the same (set to 1).

If the oscillators were decoupled, the dynamic phase $\phi_j(t)$ of oscillator j would be described by the differential equation:

$$\frac{d\phi_j}{dt} = \omega_j$$

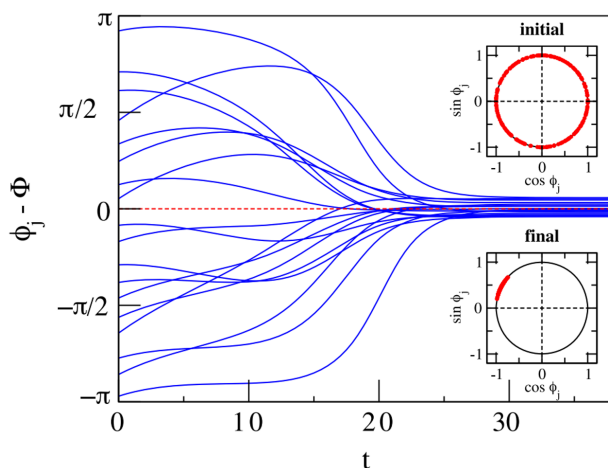
Things get more interesting however when the oscillators are coupled (*the exact mechanism that creates this coupling does not matter here*) so that the angular velocity of each oscillator depends on the phase of all other oscillators, as follows:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\phi_j - \phi_i)$$

where $K > 0$ is the coupling strength. In other words, the angular velocity of oscillator i is not just its “natural” frequency ω_i – instead the angular velocity is adjusted by the product of K and the average of the $\sin(\phi_j - \phi_i)$ terms.

This model was introduced by Yoshiki Kuramoto, and it carries his name (*Kuramoto model*). Suppose we only have $N=2$ oscillators, i and j . If the j 'th oscillator is ahead of the i 'th oscillator at time t (i.e., $\phi_j(t) > \phi_i(t)$, with $\phi_j(t) - \phi_i(t) < \pi$), then the j 'th oscillator adds a positive term in the angular velocity of the i 'th oscillator, accelerating the latter and pulling it closer to the j 'th oscillator. Similar, the i 'th oscillator adds a negative term in the angular velocity of the j 'th oscillator causing it to slow down, and also pulling it closer to the i 'th oscillator. Will the two oscillators end up synchronized, i.e., having $\phi_i(t) = \phi_j(t)$ for all time?

That depends on the magnitude of K , relative to the magnitude of the difference of the two natural frequencies $|\omega_i - \omega_j|$. Intuitively, the larger the difference of the two natural frequencies, the larger the coupling strength should be so that the two oscillators synchronize.



Actually, it is not hard to see that in this simple model the two oscillators cannot synchronize if $K < |\omega_i - \omega_j|$ because in that case, the velocity adjustment added by the other oscillator is always lower than the difference of the two natural frequencies.

What happens when we have more than two oscillators, however? To illustrate, look at these simulation results with $N=100$ oscillators.

The plot shows the phase difference between individual oscillators (*only 18 out of 100 curves are shown*) and their average phase $\Phi(t)$. In this case, K is large enough to achieve complete synchronization.

The two smaller plots inside the left figure show the initial state (*the 100 oscillators are uniformly distributed on the unit cycle of the*

complex plane) as well as the final state in which the oscillators are almost perfectly synchronized.

Let us now introduce a synchronization metric that will allow to measure how close N oscillators are to complete synchronization. Recall from calculus that a sinusoidal oscillator with phase $\phi(t)$ and unit magnitude can be represented in the complex plane as $e^{i\phi(t)}$, where i is the imaginary unit.

The average of these N complex numbers is another complex number $r(t)$ with magnitude $R(t)$ and phase $\Phi(t)$:

$$r(t) = R(t) e^{i\Phi(t)} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j(t)}$$

The magnitude $R(t)$ is referred to as **synchronization order parameter**. Its extreme values correspond to complete synchronization when $R(t)=1$ (*all oscillators have the same phase*) and complete de-coherence when $R(t)=0$ (that happens when each oscillator has a phase difference $2\pi/N$ from its closest two oscillators).

The visualization at the right shows the unit cycle in the complex plane. Two oscillators are shown as green dots on the unit cycle. The vector z represents the average of the two oscillators. The order parameter in this example is 0.853 (*shown as R in this plot*), and the phase of z is shown as $\Psi = 0.766$.

Food For Thought

Show mathematically that :

$$\frac{d\phi_i}{dt} = \omega_i + K R(t) \sin(\Phi(t) - \phi_i)$$

